

SAFIUS SAKIB SHUDDHO

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EDUCATION

BACHELOR OF SCIENCE (B.SC.) IN NUCLEAR ENGINEERING

Apr 2021 – Jul 2025

Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

CGPA: 3.49/4.00

Thesis Title: NEUTRONICS CHARACTERISTICS ANALYSIS OF ALFRED WITH LFR REFLECTOR COMPARISON AND EVALUATION OF A CONCEPTUAL MPRR USING OPENMC Jan 2024 – May 2025

Supervisor: Prof. Dr. Abdus Sattar Mollah

- Performed advanced eigenvalue (criticality) calculations using Monte Carlo neutron transport (OpenMC), validated against standard nuclear data libraries (JEFF-3.3, ENDF/B-VIII.0).
- Conducted full-core neutronics and burnup analyses of the ALFRED lead-cooled fast reactor and a conceptual multi-purpose research reactor (MPRR), achieving high precision (within 0.054%).
- Evaluated neutron flux spectra, control rod worth, reflector effects, and reactivity coefficients.
- Modeled fuel depletion, fission product poisoning, and activation for performance/safety assessment.

RESEARCH INTERESTS

- Particle Physics, Astroparticle Physics
- Physics Beyond the Standard Model, Dark Matter Physics
- Nuclear Physics, Accelerator Physics

PUBLICATIONS AND CONFERENCE PROJECTS

1. Shuddho, S. S., Sunny, A. A., & Mollah, A. S., "Neutronic Performance of Reflector Materials in Lead-cooled Fast Reactor," *Under Revision in Nuclear Science and Engineering*, 2026
2. Zarin N. Z., Sunny A. A., Shuddho, S. S. and Mollah, A. S., "Investigation of Neutronics Parameters of Multi-Purpose Research Reactor (MPRR) using VVR-KN Fuel." *International Conference on Electronics and Informatics 2024*, AEC, DHAKA, 2024.
3. Dipto, R. R., Shuddho, S. S., Sunny, A. A., & Mollah, A. S., "Analysis of Neutronics Parameters of Different Annular Fuel using Monte Carlo Code OpenMC Utilizing JEFF-3.3 and ENDF/B-VIII.0 Nuclear Data Libraries," *Proceedings of the Energy Conference 2023: National and Global Issues (ENCON23)*. Available at <http://dx.doi.org/10.2139/ssrn.4997514>.

CERTIFICATIONS

1. **Undergraduate Winter School on Particle Physics** Nov 2025
 - Completed an intensive lecture series on quarks, leptons, Higgs and neutrino physics, CP violation, sterile neutrinos, and neutrinoless double-beta decay.
 - Strengthened technical foundations in Feynman diagrams, particle accelerators, detector technologies, and modern high-energy physics experiments.
2. **Second Joint ICTP-IAEA Workshop on Open-Source Nuclear Codes for Reactor Analysis**,
Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy Sep 2025
 - Participated in hands-on sessions with OpenMC, OpenFOAM, MOOSE, and Cardinal.
 - Presented a poster titled "*Neutronic Evaluation of Reflector Materials for the ALFRED Core*".
3. **MTV Nuclear Engineering Summer School**,
Consortium for Monitoring, Technology, and Verification Jun-Aug 2023
 - Covered nuclear structure, radiation interactions, Monte Carlo methods, and nuclear data.
 - Explored applications in radiation detection, non-destructive assay, and medical physics.
4. **CS50: Introduction to Computer Science**,
Harvard University Jan-Dec 2023
 - Gained proficiency in C, Python, SQL, and web technologies (HTML/CSS, JavaScript) while applying concepts such as data structures, resource management and security.
 - Completed 10 rigorous problem sets, 9 labs, and delivered a final programming project.

TECHNICAL SKILLS

- **Computational Tools:** OpenMC, MOOSE/Cardinal (*Multiphysics*), OpenFOAM/Foam4Nuclear (*CFD*), ARMI, OpenModelica, COMSOL Multiphysics
- **Programming Languages:** Python (*NumPy, Matplotlib, Pandas*), C/C++, MATLAB, JavaScript, SQL
- **OS and Other Tools:** Linux, LaTeX, Microsoft Office, Git, GitHub

STANDARDIZED TEST SCORES

- **IELTS Academic:** Overall Band Score 7.5 [CEFR Level: C1]
Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.5.

COMPUTATIONAL PROJECTS

1. **LINACLab: Relativistic Charged-Particle Tracking and Linear Beam Optics from First Principles.**
 - Developed a framework for relativistic charged-particle tracking in EM fields, implementing integrators (Euler, RK4) and a relativistic Boris pusher validated against analytical tests.
 - Implemented beam-level optics including second-moment transport, RMS emittance and Twiss reconstruction, and Hamiltonian-derived linear maps for drifts and quadrupoles.
 - Project website: **LINACLab**; **Report Link**.
2. **DecayMC: A simple Monte Carlo framework for relativistic particle decays and scattering.**
 - Implemented Lorentz-covariant four-vector kinematics, two-body and three-body decays, Michel-weighted muon decay, and Rutherford scattering.
 - Validated against analytic benchmarks; full reproducible technical report included. **Report Link**.

SELF-STUDIES AND COURSES

1. **Quantum Mechanics** (*Introduction to Quantum Mechanics David J. Griffiths*)
2. **Electromagnetism** (*Electricity and Magnetism by Edward M. Purcell & David J. Morin*)
3. **Classical Mechanics** (*Classical Mechanics by Herbert Goldstein, Charles P. Poole & John Safko*)
4. **Mathematical Methods** (*Mathematics for Physicists by Philippe Dennery & André Krzywicki*)
5. **Fundamentals of Reactor Physics with Python**, International Atomic Energy Agency (IAEA)
 - Studied neutron slowing-down theory, multi-group transport and diffusion, point kinetics, and fuel depletion. **Books:** *Nuclear Reactor Analysis* (Duderstadt & Hamilton, 1976) and *Nuclear Reactor Physics* (Stacey, 2nd ed.).
 - Completed 11 guided labs and 8 independent Python assignments.
 - Scored **98.08%** (rank **52/638**, top 8%).

AWARDS AND ACHIEVEMENTS

Bronze Honour, International Astronomy and Astrophysics Competition (IAAC) Jul 2022

EXTRACURRICULAR ACTIVITIES

- President**, MIST Nuclear Engineering Club (MNEC) 2024 – 2025
- Established cross-year collaboration, mentoring 2nd-year students in core nuclear theory and training 3rd-year students in research tools (NumPy, Matplotlib, pandas, PyNE, OpenMC).
 - Organized **Nuclear Tempest '24**, a nuclear science outreach and competition event featuring *Nuclear Quiz* on fundamental nuclear physics and reactor concepts, *Radioactive Treasure Hunt* (radioactive source finding with GM counter as an interesting treasure hunt competition), and *Nuclear Mind Game* focused on logical and conceptual problem-solving.
- Club Coordinator**, MIST Nuclear Engineering Club (MNEC) 2023 – 2024
- Led a 14-member volunteer team in planning and executing student-led nuclear science activities and discussions.
 - Coordinated invited talks on academic motivation, undergraduate research experiences, and career pathways in nuclear science and engineering.
- Organizer**, The 4th International Conference on Energy and Power (ICEP 2022) 2022
- Designed official posters and banners used for conference promotion and branding.
 - Worked with the organizing team and volunteers to coordinate logistics and support event execution.